WEEK 9: ASSIGNMENT

1. Which of the following statement/statements are true for dual active bridge converter.
(A) It can facilitate the bidirectional power transfer
(B) It cannot facilitate the bidirectional power transfer
(C) It requires inductive filter at the output
(D) It has two active bridges

Answer: A, D

2. In a DAB operating with single phase shift modulation, at what phase shift (in degrees) $\frac{1}{5}^{th}$ of the maximum power transfer is obtained. Assume phase shift is kept less than 90 degrees. Round up your answer by 2 decimal places.

Answer: 9.486

Range: 9.00 to 10.00

3. Calculate the core cross sectional area for the transformer required in m^2, if the peak value of AC flux density is 0.1T. The primary winding flux linkage (λ) value is 400 V-sec and number of turns in the primary winding is 10. Note: The answer should be rounded up to 2 decimal places.

Answer: 200

Range: 190 - 210

4. For the transformer $P_{cu,tot} = a\left(\frac{1}{\Delta B}\right)^2$ & $P_{core} = b(\Delta B)^{\beta}$. Where a, b, and β are the coefficients. If β = 2, then the optimum value of magnetic flux density (ΔB) for the transformer is:

Note: P_{core} is the core loss; $P_{cu,tot}$ is the total copper loss;

A. $\left(-\frac{a}{b}\right)^{0.25}$

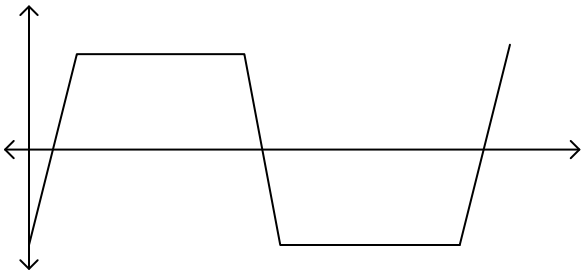
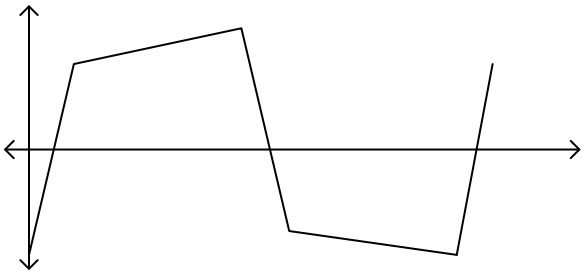
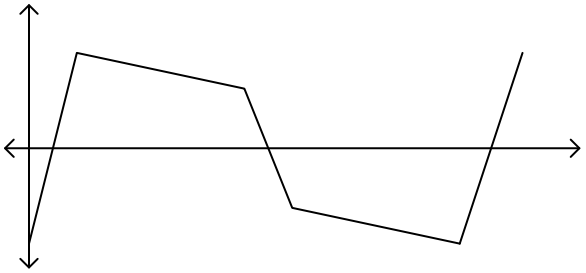
B. $\left(\frac{a}{b}\right)^{0.25}$

C. $\left(-\frac{a}{b}\right)^{0.5}$

D. $\left(\frac{a}{b}\right)^{0.5}$

Answer: B

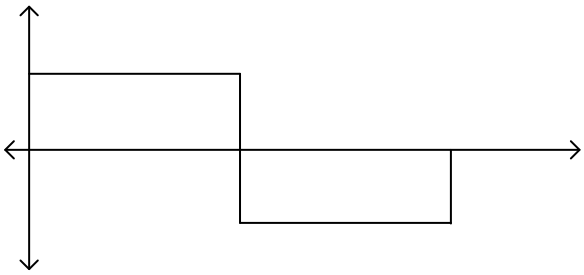
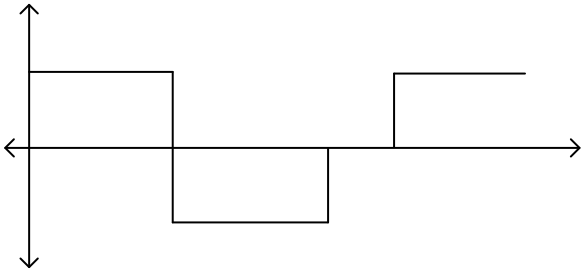
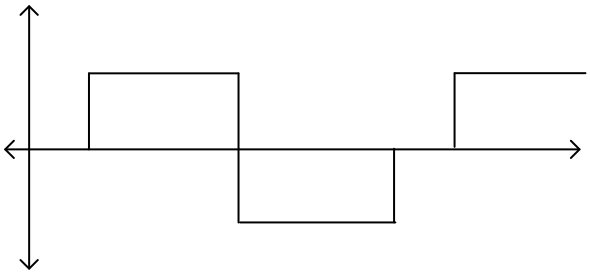
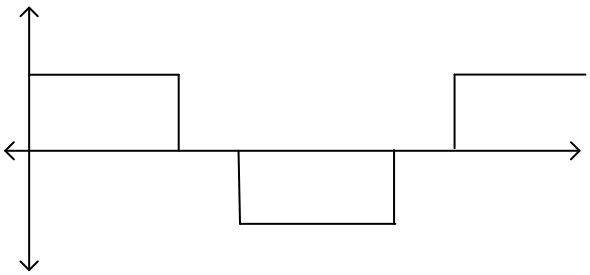
5. For the DAB operating with single phase shift match the following correct inductor current variation for corresponding input and output voltage relationship.

Input and output voltage relationship	Inductor current
I. $V_{in} > \frac{V_o}{n}$	a. 
II. $V_{in} < \frac{V_o}{n}$	b. 
III. $V_{in} = \frac{V_o}{n}$	c. 

A. I-a, II-b, III-c
B. I-b, II-c, III-a
C. I-c, II-a, III-b
D. I-b, II-a, III-c
E. I-c, II-b, III-a
F. I-a, II-c, III-b

Answer: B

6. If there is a phase shift introduced in between the half bridges of one of the full bridges of DAB, then what will the output voltage of full bridge look like.

a.	
b.	
c.	
d.	

Answer: D

7. The dual active bridge operating with single phase shift modulation having a phase shift of 60°. The output voltage is obtained as 800V with transformer having 2 turns in primary and 4 turns in secondary sides, the power transfer inductance is kept as 10 µH and the input voltage is kept at 200V. The switching frequency is maintained at 100 kHz. Then, calculate the power transfer (in kW). Note: The answer should be rounded up to 2 decimal places.

Answer: 8.89

Range: 7.50 to 10.00

8. The dual active bridge operating with single phase shift modulation achieves the power transfer with the input voltage of 400V. The output voltage is obtained as 400V with transformer having 1 turns in primary and 4 turns in secondary, power transfer inductance of 100 µH and operated with a switching frequency of 100 kHz. Then, what will be the maximum achievable power transfer (in kW).

Answer: 0.5

Range: 0.4 to 0.6

9. For a dual active bridge operated in single phase shift modulation and having 3:1 transformer ratio, if the transformer has the flat top current then:

A. $V_o = 3V_{in}$

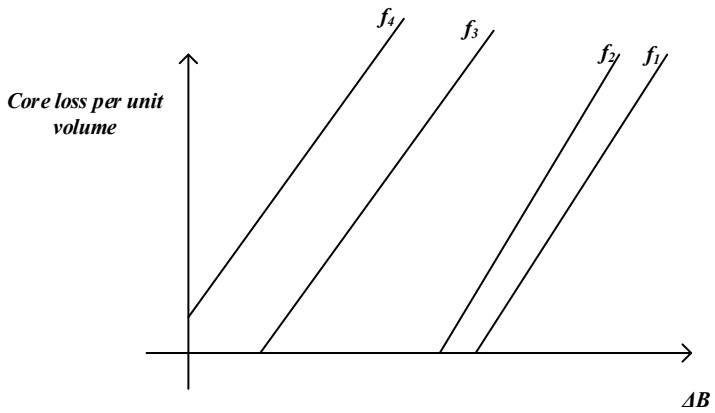
B. $V_o = \frac{V_{in}}{3}$

C. $V_o = V_{in}$

D. $V_o = 0$ and V_{in} is non-zero

Answer: B

10. Core loss per unit volume for different frequencies of excitation is shown below, then what is the minimum frequency of excitation**.**



A. f_1

B. f_2

C. f_3

D. f_4

Answer: A

Answer Key:

1. A, D	2. 9.00 to 10.00
3. 190 to 210	4. B
5. B	6. D
7. 7.50 to 10.00	8. 0.4 to 0.6
9. B	10. A